Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/18/2012
PI Name:	Ferkul, Paul Ph.D.		
Project Title:	SPICE-BASS: Smoking Point in Co-Flow Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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PI Address 2:	National Center for Space Exploration Research		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Combustion Rapid Turn Around Flight Opportunity
Start Date:	10/01/2009	**End Date:**	06/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-133 ; ISS NOTE: New end date is 6/30/2013 per PI (Ed., 6/8/2012) NOTE: project to be extended per PI (Ed., 1/19/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra (NASA Glenn Research Center) Takahashi, Fumiaki (National Center for Space Exploration Research) T'ien, James (Case Western Reserve University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The experiments described here are based on the hypothesis that materials in microgravity, with adequate ventilation, burn as well if not better than the same material in normal gravity with other conditions being identical (pressure, oxygen concentration, temperature, etc.) To test this hypothesis with any but the thinnest materials requires extended microgravity time – longer than available in drop towers or parabolic aircraft. If this hypothesis is true, it implies serious safety concerns over the utility and margin of safety provided NASA-STD-6001 Test #1. There are important differences in the suppression of fires in space compared to on Earth. On Earth it is understood that best results are generally obtained when the extinguisher "attacks" the base of the flame, which is both the stabilization point and the point of fresh air entrainment. For a fire burning in microgravity, the best point of application of suppressant may not be immediately apparent, especially for a partially obstructed flame or a wake-stabilized flame. Depending on the geometry of the flame and the characteristics of the extinguisher (distance from flame, dispersion angle) it is possible that the suppressant stream will be ineffective or might actually make the flame worse through the entrainment of oxygen. There are two main objectives: 1.) Begin to bridge the gap between the normal gravity NASA-STD-6001 Test#1 method, ground-based microgravity tests, and actual material flammability in microgravity. The relation between these can be investigated by observing flames burning for the longer times available in flight. Practical, realistic (thicker) fuels in typical geometries will be examined, including wake flames which have characteristics of a fire burning behind another object (for example, a fire burning in tight area which may not be easily accessible). 2.) Assess the effectiveness of an inert, gaseous extinguishing agent (similar to that used on ISS) in putting out flames over different materials, geometries, preheating, and flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: The current NASA spacecraft materials selection is based on a standard test method (NASA–STD–6001 Test 1) that segregates material based on 1-g behavior without consideration of low gravity effects. A critical element of this understanding is the radiative heat emission from the flame. These results are used in first order models and predictions of heat release in spacecraft fires and as a means to extend heat release data from tests like the NASA cone calorimeter test (NASA–STD–6001 Test 2) to a performance-based material selection process. Using nitrogen as a flame suppressant in microgravity provides a direct link to current and planned extinguishment techniques. Earth Applications: BASS results provide essential guidance to ground-based microgravity combustion research efforts. Detailed combustion models are validated using the simpler flow environment afforded by tests in microgravity. Once validated, they can be used to build more complex combustion models needed to capture the important details of flames burning in normal gravity. These models have wide applicability to the general understanding of many terrestrial combustion problems.
Task Progress:	The BASS hardware launched on STS-133. We reviewed still camera tests by Jay Owens and Mike Johnston burning 1-g PMMA and Delrin. The intention was to create very dim blue flames to mimic the dimmest microgravity flames expected. Camera settings were determined for these flames as the starting point for the flight experiment. Photographs of the BASS igniter wires were sent to CWRU for the purposes of the model. The geometry of the wire as well as the power as a function of time are needed to capture the ignition properly in the model. One difficulty is accounting for the relative fraction of (energy radiated)/(energy convected) from the wire. Simple heat transfer arguments show that half of the electrical energy deposited in the wire is radiated away at the typical wire temperatures. We conducted PMMA sphere combustion drop tower tests. Modeling results of sphere drop tests show qualitative agreement of the flame structure. Both the 1-g and 0-g flames as well as the transition are captured. The modeling is performed using the FLUENT software by Sam Hsu at CWRU. Results were presented at the 7th US National Technical Meeting of the Combustion Institute in March 2011. We conducted crew training with astronauts Ron Garan and Mike Fossum at JSC. While in Houston, we took the opportunity to review and modify the crew procedures with Carol Reynolds (MSFC). A description of the 1-g applications of the BASS experiment were submitted to astronaut Ron Garan. Crew training was conducted for astronaut Don Pettit. At that time, it appeared unlikely that he would perform the experiment, so science support for his crew training session was cancelled. Chuck Bunnell proceeded with training anyway so that Don is familiar with BASS. At the present time, it appears that Don Pettit will indeed perform BASS operations starting in March 2012. Crew training for Astronauts Joe Acaba (Increments 31-32) and Sunita Williams (Increments 32-33) as well as supporting MSFC operations staff was completed. We provided input to the Microgravity Science Glovebox Investigation Operations Lead for more information on the test matrix plan, including: the actual test order for each point in the test matrix; desired groupings of specific points; and an estimation of the time needed need after a group of tests for analysis before the next group. It appears that we will not have access to a dedicated combustion gas analyzer for use in the glovebox. However, Carol Reynolds (MSFC) has been working to assure that we will be able to get cabin oxygen concentration measurements before and after our tests. These are important to quantify any flame changes which might be attributed to variations in cabin oxygen. In addition, there are some fuels we prefer to burn if the cabin oxygen is on the high-end (e.g. 22%). The BASS project description was reviewed and modified, then submitted to Peter Boul (JSC) who is supporting the Office of the ISS Program Scientist. The BASS project description now appears on their "toolbox" website. In addition, the web description for the GRC ISS experiments website was updated. Procedures were reviewed in advance of the Joint Integrated Training. A request for an undergraduate student to help in the spring of 2012 has been submitted.

	We participated in Payload Expert Training to ensure proper communication techniques when operations begin. An estimate for a re-flight of BASS to burn additional fuel samples was submitted.
Bibliography Type:	Description: (Last Updated: 12/19/2013)
Papers from Meeting Proceedings	Olson SL, Johnston M, Hsu SY, T'ien JS, Ferkul PV. "Experiments and Modeling of Normal Gravity Ignition and Transition to Microgravity for a PMMA Sphere in a Forced Convective Flow." 7th US National Technical Meeting of the Combustion Institute, Atlanta, GA, March 20-23, 2011. In: 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1729-1735. Printed by Curran Associates, Inc. ISBN: 978-1-61782-780-8 , Mar-2011